ORF 409 - HW3

(Q.3) $F(x) = \frac{x+x^2}{2}, 0 \le x \le 1 \longrightarrow f_X(x) = \frac{1+2x}{2} \Rightarrow$ as $x \uparrow$, $f_X(x) \uparrow$

Inverse transform

$\Rightarrow F(x) = u \in U[0,1]$

$\Leftrightarrow x = F^{-1}$

$\Rightarrow \frac{x+x^2}{2} = u$

$\Rightarrow x^2 + x - 2u = 0$

$\Leftrightarrow x = \frac{-1 \pm \sqrt{1+8u}}{2}$

$\because x \in [0,1]$,

$x = \frac{\sqrt{1+8u} - 1}{2}$

$= F^{-1}(u)$

(Or)

$(\because u \in [0,1], 8u \in [0,8],$

$1+8u \in [1,9],$

$\sqrt{1+8u} \in [1,3]$

$\sqrt{1+8u} - 1 \in [0,2]$

$\frac{\sqrt{1+8u}-1}{2} \in [0,1]$

$\Rightarrow x \in [0,1])$

(Q.4) $F(x) = 1 - \exp(-\alpha x^{\beta}), \; 0 < x < \infty$

To generate a RV having this distribution, we use Inverse transform method again

$\Rightarrow F(x) = u \in U[0,1]$

$\Rightarrow 1 - \exp(-\alpha x^{\beta}) = u$

$\Rightarrow \exp(-\alpha x^{\beta}) = 1 - u$

$\Rightarrow -\alpha x^{\beta} = \log(1-u)$ (Here log means ln i.e. natural log)

$\Rightarrow \alpha x^{\beta} = \log\left(\frac{1}{1-u}\right)$

$\Rightarrow x^{\beta} = \frac{1}{\alpha} \log\left(\frac{1}{1-u}\right)$

$\Rightarrow x = \left[\frac{1}{\alpha} \log\left(\frac{1}{1-u}\right)\right]^{\frac{1}{\beta}}$

$\because u \in U[0,1]$,
$1-u \in U[0,1]$

$\Rightarrow x = \left[\frac{1}{\alpha} \log\left(\frac{1}{u}\right)\right]^{\frac{1}{\beta}}$

$\Rightarrow \boxed{x = \left(\frac{-\log u}{\alpha}\right)^{\frac{1}{\beta}}}$

Given u, we can find/simulate x

$F(x) \geq 0 \Rightarrow \exp(\alpha x^{\beta}) \geq 1$

$\Rightarrow \alpha x^{\beta} \geq 0$

$\Rightarrow \boxed{\alpha \geq 0}$ — (1)

$F(\infty) = 1$ ~~$\Rightarrow \beta \geq 0$ — (2)~~

$\Rightarrow \exp(-\alpha(\infty)^{\beta}) = 0$

$\Rightarrow \exp(\alpha(\infty)^{\beta}) = \infty$

$\Rightarrow \boxed{\beta \geq 0}$ — (2)

Note that $\alpha = 0 \Rightarrow F(x) = 0 \; \forall x \Rightarrow \times$

$\Rightarrow \boxed{\alpha > 0}$ — (1).

Also, $\beta = 0$

$\Rightarrow F(x) = 1 - \exp(-\alpha) < 1 \; \forall x$

$\cdot \times$

$\Rightarrow \boxed{\beta > 0}$ — (2)

(Q.6) $X \sim \exp(-x)$

$\Rightarrow f_X(x) = \exp(-x)\ ;\ x \geq 0$

To simulate ~~AR~~ r :-

where distb^n = $f_{X|X<0.05}(x)$

$\Rightarrow$ ~~f_Y~~

$$f_{X_{new}}(x) = \frac{e^{-x}}{1-e^{-0.05}},\ 0 < x < 0.05$$

$$F_{X_{new}}(x) = P(X_{new} < x),\ 0 < x < 0.05$$

$$= \int_0^x f_{X_{new}}(t)\,dt,\ 0 < x < 0.05$$

$$= \frac{1-e^{-x}}{1-e^{-0.05}},\ 0 < x < 0.05$$

Using Inverse transform method,

$$F_{X_{new}}(x) = u \in U[0,1]$$

$$\Rightarrow \frac{1-e^{-x}}{1-e^{-0.05}} = u$$

$$\Rightarrow \frac{1-e^{-x}}{c} = u \quad (\text{where } c = 1-e^{-0.05})$$

$$\Rightarrow 1-e^{-x} = cu$$

$$\Rightarrow e^{-x} = 1-cu \longrightarrow \text{natural log}$$

$$\Rightarrow -x = \log(1-cu)$$

$$\Rightarrow \boxed{x = \log\left(\frac{1}{1-cu}\right) = F^{-1}(u)}$$

$E(X \mid X < 0.05)$ $\quad e^{-0.05} = 0.9512$

$$= \int_0^{0.05} \frac{x\,e^{-x}}{1-e^{-0.05}}\,dx$$

$$= \frac{\left(xe^{-x}\right)_{0.05}^{0} + \int_0^{0.05} e^{-x}\,dx}{1-e^{-0.05}}$$

$$= \frac{-0.05e^{-0.05} + 1 - e^{-0.05}}{1-e^{-0.05}}$$

$$= \frac{1 - 1.05e^{-0.05}}{1-e^{-0.05}}$$

$$= \frac{1 - 1.05 \times 0.95}{1-0.95}$$

$$= \boxed{0.0248}$$

If we simulate $f_{X_{new}}(x)$ by rejection method,

where $Y = U[0, 0.05]$

$\Rightarrow f_Y(y) = 20$

$$\Rightarrow \frac{f_{X_{new}}(x)}{f_Y(y)} = \frac{e^{-x}}{20(1-e^{-0.05})} = h(x)$$

max $h(x)$ when $x = 0$

$$= \frac{1}{20 \times 0.05} = \frac{1}{<1} = >1$$

$\Rightarrow$ (# trials) needed through rejection method = max $h(x) > 1$

$\Rightarrow$ # trials by Inverse transform method $= 1$

$\Rightarrow \because$ Inverse transform method needs less # trials, it is more efficient

(Q.12)

(a) Consider the variable :-

$X = \max(X_1, X_2, \ldots, X_n)$

$F_X(x) = P(X \le x)$

$= P(\max(X_1, X_2, \ldots, X_n) \le x)$

$= P(X_1 \le x, X_2 \le x, \ldots, X_n \le x)$

$= P(X_1 \le x)\, P(X_2 \le x) \ldots P(X_n \le x)$

$\because X_i \perp\!\!\!\perp X_j \ \forall i \ne j$

$= F_{X_1}(x)\, F_{X_2}(x) \ldots F_{X_n}(x)$

$= \prod_{i=1}^{n} F_{X_i}(x)$

So, given it is easy to generate RV from the distbn F_i, generate them $\Rightarrow X_i\ (X_1, X_2, \ldots)$

Just take the max of these X_i's

i.e. $X = \max(X_1, \ldots, X_n)$

$\Rightarrow$ we have generated X i.e. the RV that follows the distbn

$F_X(x) = \prod_{i=1}^{n} F_{X_i}(x)$

$\Rightarrow$ To generate a RV that follows (X), just take max of $(X_1, X_2, \ldots, X_n)$

(b) Consider the variable :-

$X = \min(X_1, X_2, \ldots, X_n)$

$F_X(x) = P(X \le x)$

$= 1 - P(X > x)$

$= 1 - P(\min(X_1, X_2, \ldots, X_n) > x)$

$= 1 - P(X_1 > x, X_2 > x, \ldots, X_n > x)$

$= 1 - P(X_1 > x)\, P(X_2 > x) \ldots P(X_n > x)$

$(\because X_i \perp\!\!\!\perp X_j \ \forall i \ne j)$

$= 1 - \prod_{i=1}^{n} P(X_i > x)$

$= 1 - \prod_{i=1}^{n} (1 - P(X_i \le x))$

$= 1 - \prod_{i=1}^{n} (1 - F_{X_i}(x))$

So, given it is easy to generate RV (X_i) from F_i, generate them. Then, take their min

$\Rightarrow X = \min(X_1, X_2, \ldots, X_n)$

$\Rightarrow$ we have generated a RV X that follows :-

$F_X(x) = 1 - \prod_{i=1}^{n} (1 - F_{X_i}(x))$

(Q.24) Let $A = |Z|$ $\in N(0,1)$

$$\Rightarrow f_A(a) = \frac{2}{\sqrt{2\pi}} e^{-a^2/2}$$

$$g(x) = \lambda e^{-\lambda x}$$

let $f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}$

Now, $\frac{f(x)}{g(x)} = \frac{\frac{2}{\sqrt{2\pi}} e^{-x^2/2}}{\lambda e^{-\lambda x}}$

$$= \frac{2}{\lambda\sqrt{2\pi}} \exp\left(\lambda x - \frac{x^2}{2}\right)$$

$= \max$ when $\lambda x - \frac{x^2}{2} = \max$

$= \max$ when $\boxed{\lambda = x}$

$$c = \frac{f(x)}{g(x)}\Big|_{x=\lambda} = \frac{2}{\lambda\sqrt{2\pi}} \exp\left(\lambda^2 - \frac{\lambda^2}{2}\right)$$

$$= \frac{2}{\lambda\sqrt{2\pi}} \exp\left(\frac{\lambda^2}{2}\right)$$

$\max \frac{f(x)}{g(x)}$

Now, c is the mean # iterations required in the rejection method.

To minimize # iterations, we need to minimize c (wrt λ),

$$\Rightarrow \frac{dc}{d\lambda} = 0 \Rightarrow \frac{2\lambda\left(\lambda \exp\left(\frac{\lambda^2}{2}\right)\right) - \exp\left(\frac{\lambda^2}{2}\right)}{\sqrt{2\pi}\,\lambda^2} = 0$$

$$\Rightarrow \lambda^2 = 1$$

$$\Rightarrow \lambda = \pm 1$$

$\because g(x)$ is a density f^n,

$g(x) > 0$

$\lambda e^{-\lambda x} > 0$

$\Rightarrow \boxed{\lambda = 1}$

To further check whether $\lambda = 1$ ~~extrem~~ extremum point is a point of minima we compute $\frac{d^2c}{d\lambda^2}$ & check if it is ≥ 0

So, $\frac{d^2c}{d\lambda^2} = \frac{d}{d\lambda}\left(\frac{dc}{d\lambda}\right)$

$$= \frac{d}{d\lambda}\left(\underbrace{\frac{2}{\sqrt{2\pi}}}_{c_0}\frac{(\lambda^2-1)}{\lambda^2}\exp\left(\frac{\lambda^2}{2}\right)\right)$$

$$= c_0\left[\left(\frac{\lambda^2-1}{\lambda^2}\right)\lambda\exp\left(\frac{\lambda^2}{2}\right) + \exp\left(\frac{\lambda^2}{2}\right)\frac{2}{\lambda^3}\right]$$

$$= c_0\exp\left(\frac{\lambda^2}{2}\right)\left(\lambda - \frac{1}{\lambda} + \frac{2}{\lambda^3}\right)\Big|_{\lambda=1}$$

$$= c_0\exp\left(\frac{1}{2}\right)\left(1 - \frac{1}{1} + \frac{2}{1}\right)$$

$$> 0$$

$\Rightarrow \boxed{\lambda = 1}$ is indeed a pt of minimum of c

$\Rightarrow c = c(\lambda)$ is minimized when $\lambda = 1$.

(Q.26) Given:-

(X, Y) uniformly distributed in a circle of radius 1

$R \geq 0 \Leftarrow$

So, $P(R^2 \leq \theta) = \cancel{P(R \leq \sqrt{\theta})}$

$= P(R \leq \sqrt{\theta})$

$=$ # points (in the o/e) at a distance $\leq \sqrt{\theta}$ from the centre

~~Total~~ # points ~~(in the o/e)~~ ~~entire region~~

($\because$ ~~# points~~ this setting is a 2D setting, (*)

points in a region $\propto$ area of the region)

disc $\Rightarrow$ pts can be inside the rim also

$\Rightarrow P = \dfrac{\cancel{\pi\sqrt{\theta}}\ \text{\# points in the disc of radius } \sqrt{\theta}}{\text{\# points} \quad // \quad 1}$

using (*) $= \dfrac{\pi(\sqrt{\theta})^2}{\pi(1)^2}$

$= \dfrac{\cancel{\pi}\theta}{\cancel{\pi}}$

$= \theta$

$\Rightarrow P(R^2 \leq \theta) = \theta \; ; \; \theta \in [0,1]$

$\Rightarrow F_{R^2}(\theta) = \theta \; ; \; \theta \in [0,1]$

$\Rightarrow f_{R^2}(\theta) = 1 \; ; \; \theta \in [0,1]$

$\Rightarrow f_{R^2}(\theta) = \begin{cases} 0 & ; \theta < 0 \\ 1 & ; 0 \leq \theta \leq 1 \\ 0 & ; 1 < \theta \end{cases}$ $\Rightarrow$ R^2 is uniform on $[0,1]$

= QED

```
import math
import numpy as np
```

Chapter 5

Q3

```
def F_inv(u):
    return (math.sqrt(1+8*u)-1)/2

u = np.random.rand()
x = F_inv(u)
print(u, x)
```

```
0.6141577507857493 0.7158599843614801
```

Q4

```
def F_inv(u, alpha, beta):
    return (-math.log(u)/alpha)**(1/beta)
```

Q6

```
def F_inv(u):
    c = 1 - math.exp(-0.05)
    return math.log(1/(1-c*u))

u = np.random.rand()
x = F_inv(u)
print(u, x)
```

```
0.121170154547935 0.005927068589909094
```

```
ans = 0
num_trials = 1000
for _ in range(num_trials):
    u = np.random.rand()
    x = F_inv(u)
    ans += x
ans = ans/num_trials
print(ans)
```

```
0.024679283005896988
```

Q19

a

```
def F_inv(u):
    return (math.sqrt(1+8*u)-1)/2
```

```
u = np.random.rand()
x = F_inv(u)
print(u, x)
```

```
0.12588268203018382 0.2083539821730147
```

#trials needed = 1

b

```
while(True):
    u1 = np.random.rand()
    u2 = np.random.rand()
    if(u2<=(1+2*u1)/3):
        x = u2
        break
print(x)
```

```
0.7249164539711337
```

#trials needed = c = max(pi/qi) = max((1+2*x)/2/1) = 3/2

c

```
u = np.random.rand()
if(u<0.5):
    u1 = np.random.rand()
    x = u1
else:
    u2 = np.random.rand()
    x = math.sqrt(u2)
print(x)
```

```
0.8608716749018118
```

#trials needed = 3

Since the number of trials needed is the least in inverse transform method, it is the most efficient option. Hence, we go with that!

Q25

```
while(True):
    u1 = np.random.rand()
    u2 = np.random.rand()
    y = -math.log(u2)
    if(u1<=math.exp(-(y-1)**2/2)):
        x = y
        break
output = 0
```

```
u3 = np.random.rand()
if(u3<=0.5):
    output = x
else:
    output = -x
print(output)
```

```
-1.6279386991917928
```